

10.3 Polynomial Regression



Core skills covered in this section:

- ▶ Model nonlinear relationships by artificially constructing higher-power features.
- ▶ The design matrix for polynomial regression, and the expanded vector space.
- ▶ Using the least-squares formula to compute polynomial regression coefficients.
- ▶ Understanding underfitting and overfitting.
- ▶ Bivariate and even multivariate polynomial regression, mastering cross terms and higher-order combinations.

Nonlinear Features

In multivariate linear regression discussed earlier in this chapter, the independent variables $x_1, x_2, \dots, x_D$ are all original features that can be obtained through actual measurement — such as height, weight, and age.

But in practice, modeling isn't always limited to these original linear features. In particular, when a single feature isn't enough to capture a complex nonlinear relationship (as shown in Figure 1), we can "manufacture" new features in a specific way. The polynomial regression introduced in this section is exactly this: adding "higher-power" nonlinear features to the regression.

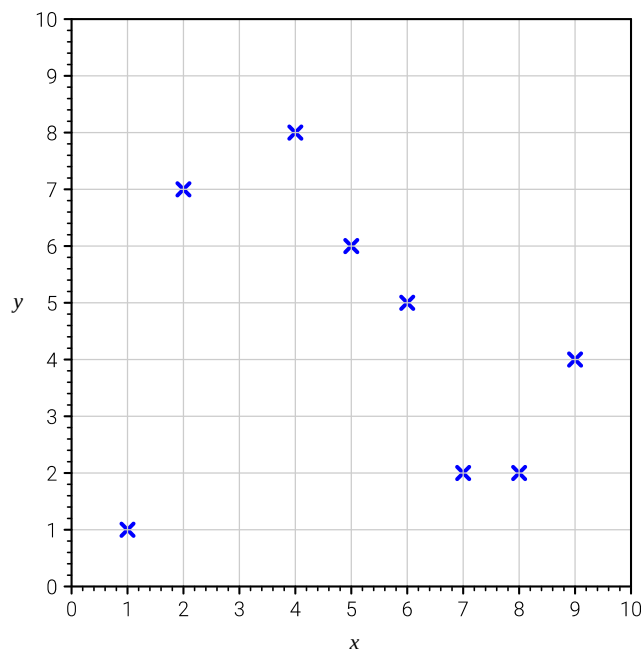


Figure 1. Scatter Data Showing a "Nonlinear" Relationship

Polynomial Regression

As Figure 2 shows, although there is only one feature x , we can artificially construct nonlinear features such as x^2 , x^3 , thereby expanding to a set of features such as $1, x, x^2, x^3$, and so on. This approach is exactly what gives rise to polynomial regression.

Univariate polynomial regression is a regression model that fits nonlinear data relationships by introducing higher powers of the original variable x (x^2, x^3 , etc.). The explicit expression for an m -th-order polynomial regression is

$$\hat{y} = b_0 + b_1x + b_2x^2 + \dots + b_mx^m$$

Here, $b_2x^2 + \dots + b_mx^m$ are all "artificial" features.

In the equation above, when m is set to 1, we obtain univariate linear regression. That is, univariate linear regression can be viewed as a special case of univariate polynomial regression.

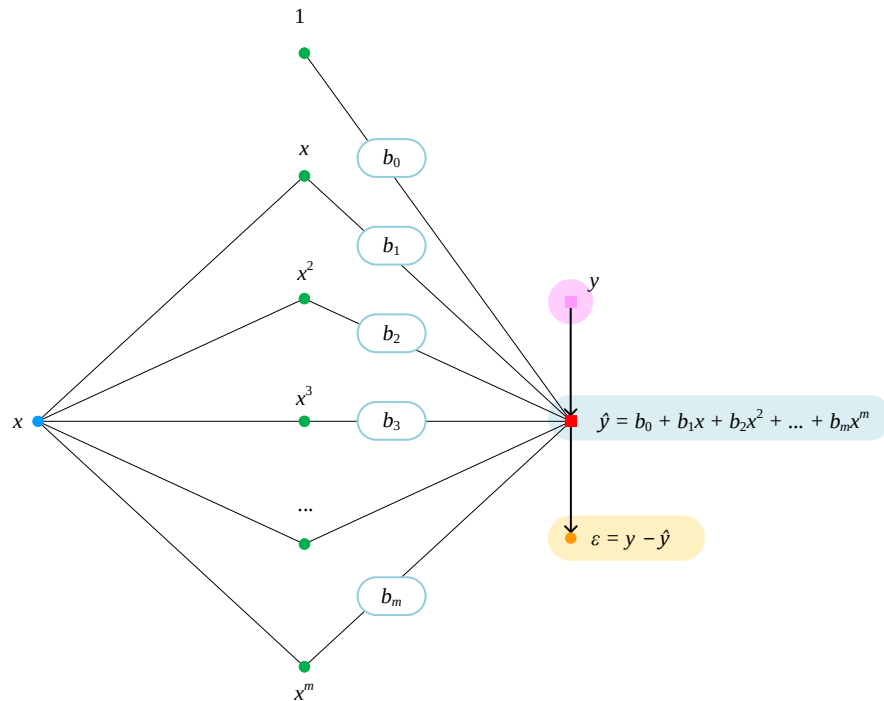


Figure 2. Data Relationship for Univariate Polynomial Regression

Although the name "polynomial regression" contains the word "polynomial," it is, in essence, still linear regression — because it fits these nonlinearly constructed new features in a linear fashion.

For example, suppose we have only one variable x . If we want to perform a quadratic (degree-2) polynomial regression, we would use three features: the constant term 1 , the linear term x , and the quadratic term x^2 , and then fit the model.

Increasing the degree of the polynomial makes the model more flexible and able to fit more complex data, but too high a degree can also easily lead to overfitting — the model performs well on the training data but poorly on new data. We will discuss this in detail at the end of this section.

Design Matrix

From a data perspective, as Figure 3 shows, in polynomial regression we use not only the original value of the independent variable but also its various powers as additional features, allowing us to better fit the nonlinear features in the data.

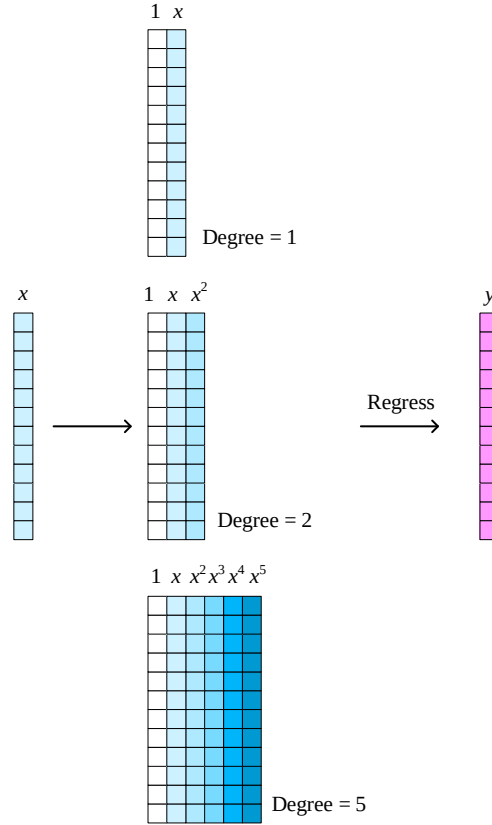


Figure 3. Constructing the Design Matrix for Univariate Polynomial Regression

For example, the design matrix X for a univariate 5th-order polynomial regression is

$$X = \begin{bmatrix} \mathbf{1} & \mathbf{x} & \mathbf{x} \odot \mathbf{x} & \cdots & \mathbf{x} \odot \mathbf{x} \odot \mathbf{x} \odot \mathbf{x} \odot \mathbf{x} \end{bmatrix} = \begin{bmatrix} 1 & x^{(1)} & (x^{(1)})^2 & \cdots & (x^{(1)})^5 \\ 1 & x^{(2)} & (x^{(2)})^2 & \cdots & (x^{(2)})^5 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x^{(n)} & (x^{(n)})^2 & \cdots & (x^{(n)})^5 \end{bmatrix}$$

Here, $\mathbf{x} \odot \mathbf{x}$ represents the element-wise product (also called the Hadamard product). The element-wise product takes two matrices of the same shape as input and produces a matrix of the same shape, where each entry equals the product of the corresponding entries in the two input matrices.

Note that, as mentioned earlier in this book, a vector is a special case of a matrix.

In this way, y can be written as

$$\begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}_y = \underbrace{\begin{bmatrix} 1 & x^{(1)} & (x^{(1)})^2 & \cdots & (x^{(1)})^5 \\ 1 & x^{(2)} & (x^{(2)})^2 & \cdots & (x^{(2)})^5 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x^{(n)} & (x^{(n)})^2 & \cdots & (x^{(n)})^5 \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_5 \end{bmatrix}}_{\mathbf{b}} + \underbrace{\begin{bmatrix} \varepsilon^{(1)} \\ \varepsilon^{(2)} \\ \vdots \\ \varepsilon^{(n)} \end{bmatrix}}_{\boldsymbol{\varepsilon}}$$

From the perspective of function graphs, as Figure 4 shows, a univariate polynomial regression model is the result of superimposing several curves.

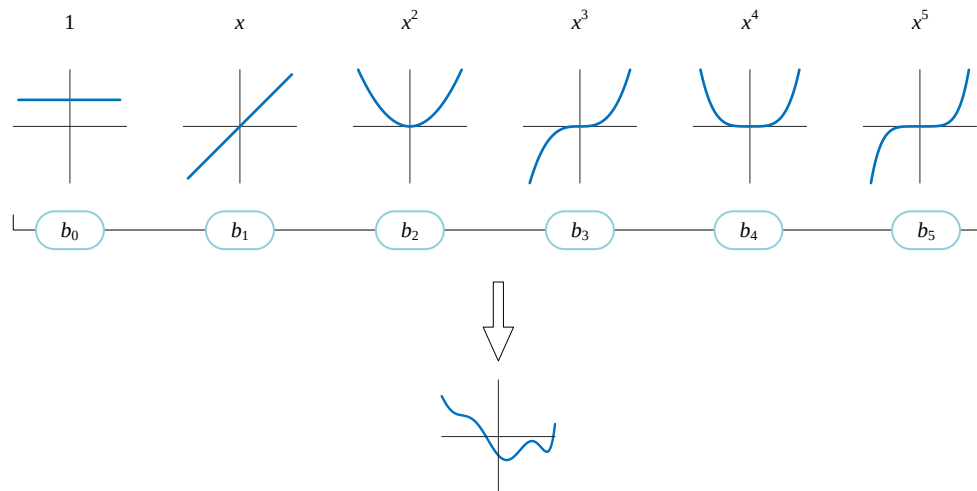
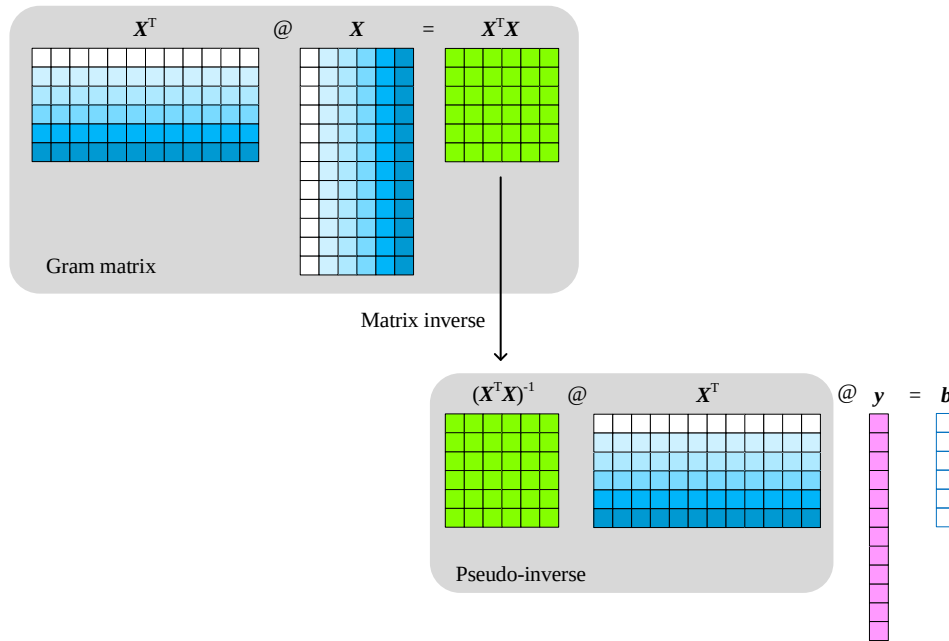


Figure 4. A Univariate Degree-5 Function Can Be Viewed as the Superposition of 6 Curves (image from Machine Learning)

Solving Polynomial Regression

The method for solving the parameters in a univariate (x)-order polynomial regression, $\mathbf{b}$, is identical to the method used earlier in this chapter for multivariate linear regression, as Figure 5 shows. In univariate polynomial regression, the column vectors of the design matrix $\mathbf{X}$ likewise span a multi-dimensional vector space.

Figure 5. Computing b , Univariate Polynomial Regression

Let's look at a concrete example.

Univariate Quadratic Polynomial Regression

Figure 1's scatter data has independent-variable column vector x and dependent-variable column vector y , respectively:

$$x = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 7 \\ 8 \\ 6 \\ 5 \\ 2 \\ 2 \\ 4 \end{bmatrix}$$

For example, the model corresponding to a degree-2 univariate (x) polynomial regression (univariate quadratic polynomial regression) is

$$\hat{y} = b_0 + b_1 x + b_2 x^2$$

In polynomial regression analysis, "quadratic" refers to the fact that the squared term of the variable is included in the model – that is, besides the constant term b_0 and the linear term $b_1 x$, the model also includes $b_2 x^2$. In the equation above, the highest degree among all the terms is called the degree of the polynomial – here, 2.

The design matrix X for univariate quadratic polynomial regression is

$$X = [\mathbf{1} \quad \mathbf{x} \quad \mathbf{x} \odot \mathbf{x}] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \end{bmatrix}$$

As Figure 6 shows, the result of orthogonally projecting vector $\mathbf{y}$ onto $\text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x} \odot \mathbf{x})$ is $\hat{\mathbf{y}}$.

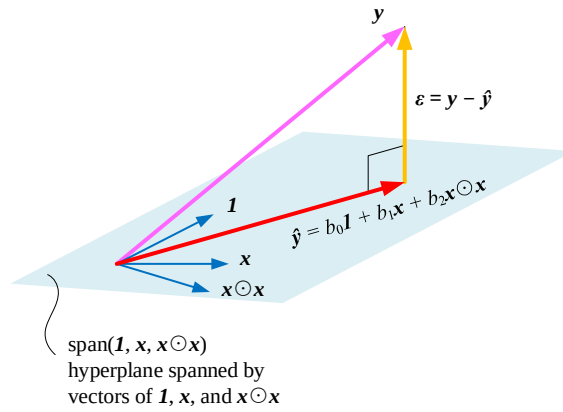


Figure 6. $\mathbf{y}$ Projected onto $\text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x} \odot \mathbf{x})$, Univariate Quadratic Polynomial Regression

Using the same method as in multivariate linear regression, we compute $\mathbf{b}$

$$\begin{aligned} \mathbf{b} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \\ &= \begin{pmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \end{bmatrix}^T \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \end{bmatrix} \end{pmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \end{bmatrix}^T \begin{bmatrix} 1 \\ 7 \\ 8 \\ 6 \\ 5 \\ 2 \\ 2 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 42 & 276 \\ 42 & 276 & 1998 \\ 276 & 1998 & 15252 \end{bmatrix}^{-1} \begin{bmatrix} 35 \\ 173 \\ 1037 \end{bmatrix} \\ &= \begin{bmatrix} 1.636 & -0.670 & 0.058 \\ -0.670 & 0.344 & -0.033 \\ 0.058 & -0.033 & 0.003 \end{bmatrix} \begin{bmatrix} 35 \\ 173 \\ 1037 \end{bmatrix} \\ &= \begin{bmatrix} 1.655 \\ 1.926 \\ -0.214 \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} \end{aligned}$$

In this way, we obtain the following univariate degree-2 polynomial regression model

$$\hat{y} = 1.655 + 1.927x - 0.214x^2$$

Figure 7's red parabola corresponds to the equation above.

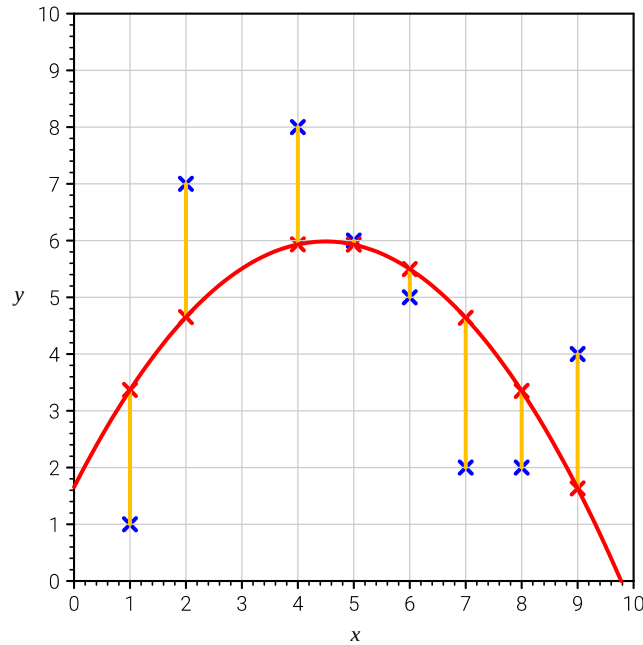


Figure 7. Degree-2 Univariate Polynomial Regression

Figure 7's orange line segments represent the error, corresponding to vector $\boldsymbol{\varepsilon}$

$$\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -2.368 \\ 2.349 \\ 2.068 \\ 0.070 \\ -0.498 \\ -2.638 \\ -1.35 \\ 2.367 \end{bmatrix}$$

The sum of squared errors is

$$\|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = \begin{bmatrix} -2.368 \\ 2.349 \\ 2.068 \\ 0.070 \\ -0.498 \\ -2.638 \\ -1.35 \\ 2.367 \end{bmatrix}^T \begin{bmatrix} -2.368 \\ 2.349 \\ 2.068 \\ 0.070 \\ -0.498 \\ -2.638 \\ -1.35 \\ 2.367 \end{bmatrix} = 30.040$$

Univariate Cubic Polynomial Regression

A lower degree may not capture the complex relationships in the data well. For example, in Figure 7, the orange line segments are still quite long.

Let's now raise the degree to 3!

The model corresponding to a degree-3 univariate (x) polynomial regression (univariate cubic polynomial regression) is

$$\hat{y} = b_0 + b_1x + b_2x^2 + b_3x^3$$

? Please write out the design matrix $\mathbf{X}$ for univariate cubic polynomial regression on your own, and compute $\mathbf{b}$.

Figure 8's red curve is exactly the univariate cubic polynomial regression model, corresponding to the equation

$$\hat{y} = -8.539 + 12.211x - 2.643x^2 + 0.16x^3$$

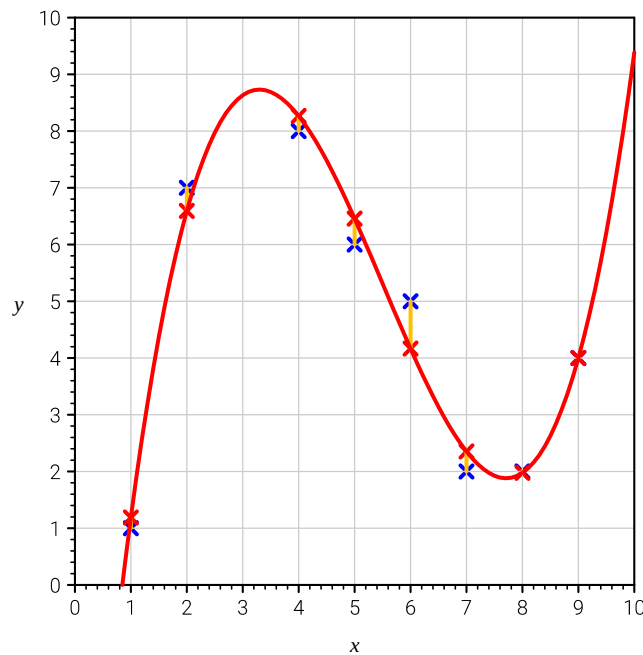


Figure 8. Degree-3 Univariate Polynomial Regression

Figure 8's orange line segments represent the error, corresponding to vector $\boldsymbol{\varepsilon}$

$$\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -0.189 \\ 0.407 \\ -0.266 \\ -0.457 \\ 0.834 \\ -0.352 \\ 0.023 \\ -0.001 \end{bmatrix}$$

The sum of squared errors is

$$\|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = \begin{bmatrix} -0.189 \\ 0.407 \\ -0.266 \\ -0.457 \\ 0.834 \\ -0.352 \\ 0.023 \\ -0.001 \end{bmatrix}^T \begin{bmatrix} -0.189 \\ 0.407 \\ -0.266 \\ -0.457 \\ 0.834 \\ -0.352 \\ 0.023 \\ -0.001 \end{bmatrix} = 1.302$$

Compared with quadratic polynomial regression, the error of cubic polynomial regression is noticeably smaller.

Overfitting

However, too high a degree can lead to overfitting. Overfitting refers to the phenomenon where a model performs well on the training data but poorly on new data. When the degree of a polynomial regression is too high, the model may over-adapt to the noise and fine details in the training data, thereby losing its generalization capability.

This means the model may be unable to make accurate predictions on new, unseen data, because it has "memorized" many minor fluctuations in the training data — fluctuations that may not actually exist in real data.

Take learning linear algebra as an example: underfitting is like being "academically unpolished" — a student has only a superficial grasp of the material, may not even have a clear understanding of matrix multiplication, and often gets basic questions wrong. This shows that the student has studied too little, too shallowly, and hasn't practiced enough to master the fundamentals of linear algebra.

Overfitting, on the other hand, is like "rote memorization" — a student memorizes the solution techniques for every example problem, but as soon as the problem type changes slightly, or requires combining textbook knowledge with practical application, they're at a loss. This shows that the learning was too mechanical and too dependent on memorized details, lacking the distillation and generalization needed to apply what was learned to new situations.

A truly good learning state is like a model that performs well on both the training set (example problems) and the test set (new problems, practical applications) — one that has grasped the underlying patterns while also gaining the flexibility to adapt.

Figure 9 shows the result of a degree-8 univariate polynomial regression; the corresponding equation is

$$\hat{y} = 7.123 - 11.430x - 2.828x^2 + 15.342x^3 - 9.463x^4 + 2.599x^5 - 0.370x^6 + 0.027x^7 - 0.001x^8$$

As Figure 9 shows, the red regression curve "perfectly" passes through all the data points, meaning the error is almost 0!

However, this is not actually an ideal model. The higher the degree, the better a polynomial regression model can fit the training data — but the more likely it is to perform poorly on test data or in real-world applications.

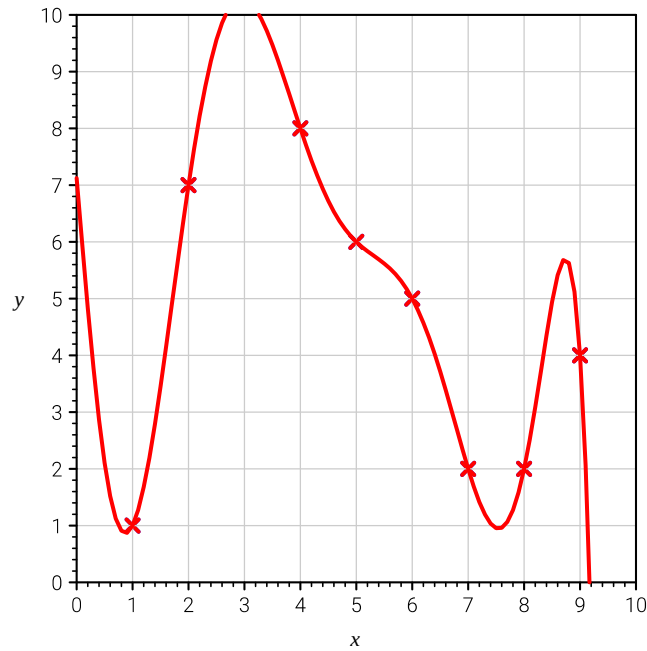


Figure 9. Degree-8 Univariate Polynomial Regression



LA_10_03_01.ipynb solves a degree-3 univariate polynomial regression and plots Figure 8.

LA_10_03_02.ipynb, on the other hand, uses the LinearRegression tool provided by Scikit-learn to complete model training and prediction, and plots Figure 9.

Bivariate Polynomial Regression

Given that univariate polynomial regression exists, it's only natural that bivariate – and even multivariate – polynomial regression exist as well.

As discussed earlier, univariate quadratic polynomial regression introduces, into the linear regression model $\hat{y} = b_0 + b_1x$, a higher-order term such as b_1x^2 . Now, if we expand the variables to two – x_1, x_2 – we can construct bivariate polynomial regression.

For example, in bivariate quadratic polynomial regression, besides the original x_1 and x_2 , we also introduce the combinations x_1^2, x_2^2 , and the cross term x_1x_2 .

Therefore, the form of the bivariate quadratic polynomial regression model is

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_1^2 + b_4x_2^2 + b_5x_1x_2$$

The equation above is equivalent to the superposition of the plots in Figure 10.

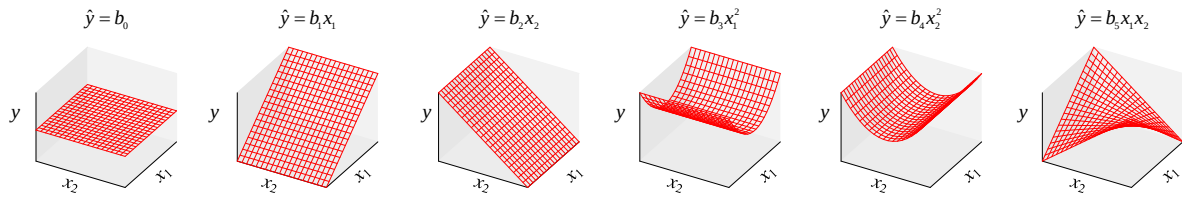


Figure 10. Plots of Each Term in the Bivariate Quadratic Polynomial

The equation above can be written as

$$\hat{y} = \begin{bmatrix} 1 & x_1 & x_2 & x_1^2 & x_2^2 & x_1 x_2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

This means the design matrix $\mathbf{X}$ for bivariate quadratic polynomial regression is

$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_1 \odot \mathbf{x}_1 & \mathbf{x}_2 \odot \mathbf{x}_2 & \mathbf{x}_1 \odot \mathbf{x}_2 \end{bmatrix}$$

This model can fit not only the linear relationship between the two independent variables but also capture the nonlinear interaction between them.

If we continue expanding upward — say, to bivariate cubic polynomial regression — even more features get constructed, including cubic terms. As Figure 11 shows, as the degree increases, we keep incorporating more higher-order terms. Through these artificially constructed features, the model can better fit complex, surface-shaped data distributions.

	1	x_2	x_2^2	x_2^3	...
1	1	x_2	x_2^2	x_2^3	
x_1	x_1	$x_1 x_2$	$x_1 x_2^2$	$x_1 x_2^3$	
x_1^2	x_1^2	$x_1^2 x_2$	$x_1^2 x_2^2$	$x_1^2 x_2^3$	
x_1^3	x_1^3	$x_1^3 x_2$	$x_1^3 x_2^2$	$x_1^3 x_2^3$	
$\vdots$					

Figure 11. Terms of Different Degrees in the Polynomial

In addition, the quadratic-term portion of the equation above can be written in matrix multiplication form

$$b_3 x_1^2 + b_4 x_2^2 + b_5 x_1 x_2 = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} b_3 & b_5/2 \\ b_5/2 & b_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{x}^T \begin{bmatrix} b_3 & b_5/2 \\ b_5/2 & b_4 \end{bmatrix} \mathbf{x}$$

This is exactly the quadratic form, which we'll introduce later in this book.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises in this section.

Q1. The following code generates the independent-variable and dependent-variable data for univariate regression; please write Python code to compute the following for a univariate degree-3 polynomial regression: $\mathbf{b}$, $\hat{\mathbf{y}}$, and ε .

```
## 初始化
import numpy as np

## 创建数据
np.random.seed(0)
n_samples = 100

# 自变量数据
x = np.random.uniform(0, 4, num)

# 因变量数据
y = np.sin(0.4*np.pi * X) + 0.4 * np.random.randn(num)
```

Q2. Use PolynomialFeatures from the sklearn.preprocessing module, along with LinearRegression from the sklearn.linear_model module, to solve the univariate degree-3 polynomial regression from Q1, and compare the results.